Thermodynamics

2/15/2019

1. (40 pts) The bomb calorimetry can be used to determine the calorific value of food.

   (a) In an experiment, the heat produced by the total combustion of a 1.00 g (3.00×10^{-3} mole) sample of chicken fat using bomb calorimetry was found to be 10,000 cal at 37 °C. Calculate the calorific value of the chicken fat in $cal \cdot g^{-1}$ at 37°C under constant pressure; that is, calculate the standard enthalpy of combustion of the chick fat at 37 °C. The reaction may be represented by the equation,
   $C_{20}H_{32}O_2(s) + 27O_2(g) \rightarrow 20CO_2(g) + 16H_2O(l)$

   (b) A student in a laboratory made a statement: "$\Delta H = \Delta E + P\Delta V$. Since the bomb calorimetry process is constant volume one, $\Delta V = 0$, and $\Delta H = \Delta E$." Is this statement correct? Explain.

2. (60 pts) The SO_2 molecule is nonlinear.

   (a) Find the molar heat capacity $\overline{C}_v$ of gaseous SO_2, as predicted by applying classical mechanics to all motions of the nuclei.

   (b) The vibrational frequencies of SO_2 are 1.57×10^{13}, 3.45×10^{13}, and 4.08×10^{13} sec^{-1}. What is the lowest temperature (roughly) at which you would expect $\overline{C}_v$ for gaseous SO_2 to agree with the result of (a)? Explain.

   (c) Consider SO_2 existing only in a hypothetical two-dimensional space (for example, the plane of this paper). What is the number of modes of vibration for SO_2? Would the result be different for CO_2? Explain.